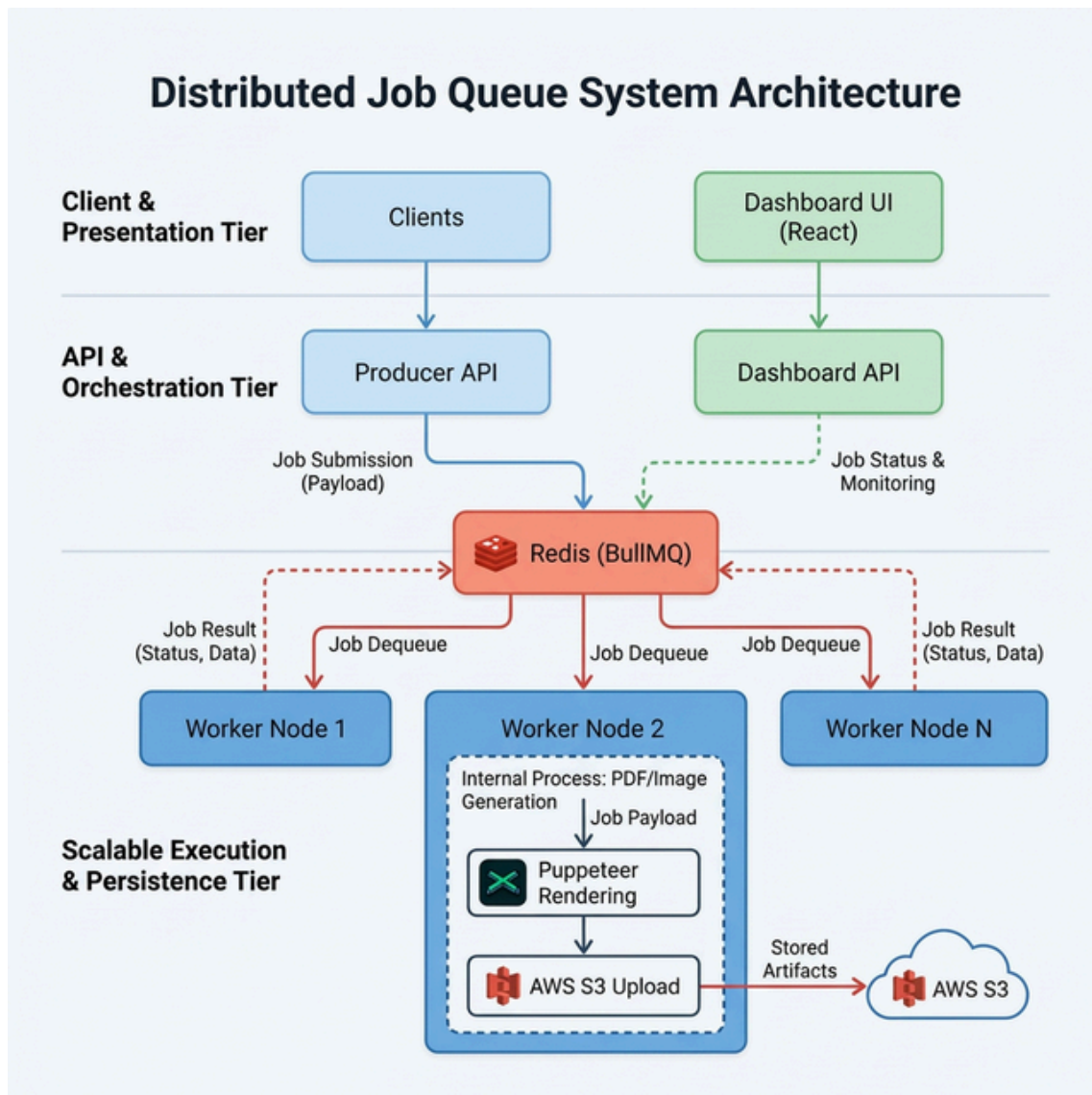


Portfolio: Distributed Job Queue System

GitHub Repository: <https://github.com/Protik111/distributed-report-queue>

1. Executive Summary

A production-grade distributed job queue system designed for asynchronous report generation. Leveraging a decoupled microservices architecture, it ensures high availability, scalability, and reliability by using Redis as a message broker and AWS S3 for persistent storage.



2. Architecture & System Design

Key Components

- **Producer:** Acts as the gateway. It validates input and pushes a "generate-report" job into the Redis queue.

- **Worker:** The "heavy lifter". Multiple replicas run concurrently to drain the queue. Processes HTML and generates PDFs via Puppeteer.
- **Scheduler:** A lightweight monitor ensuring jobs don't hang indefinitely if a worker crashes.
- **Dashboard UI & API:** Real-time queue analytics and system health monitoring UI.

Technical Stack

- **Languages/Frameworks:** Node.js, Express.js, React + Vite, TypeScript
- **Infrastructure:** Docker, AWS (S3, EC2), Redis
- **Message Broker:** BullMQ
- **IaC & CI/CD:** Pulumi (TypeScript), GitHub Actions

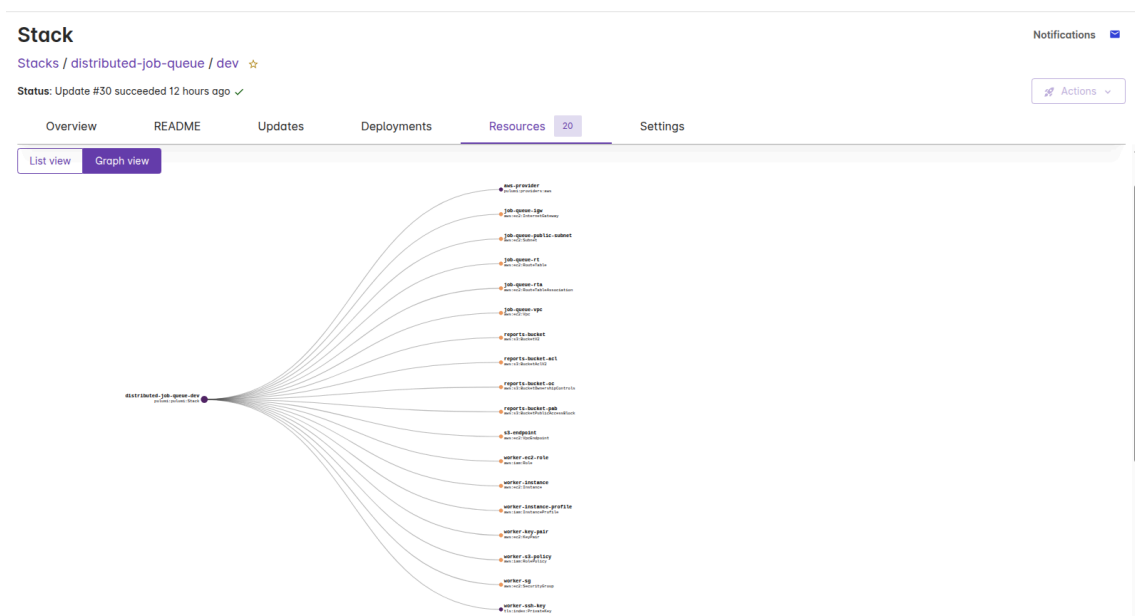
3. Worker Internals & PDF Generation

The Worker Service manages intensive PDF generation tasks:

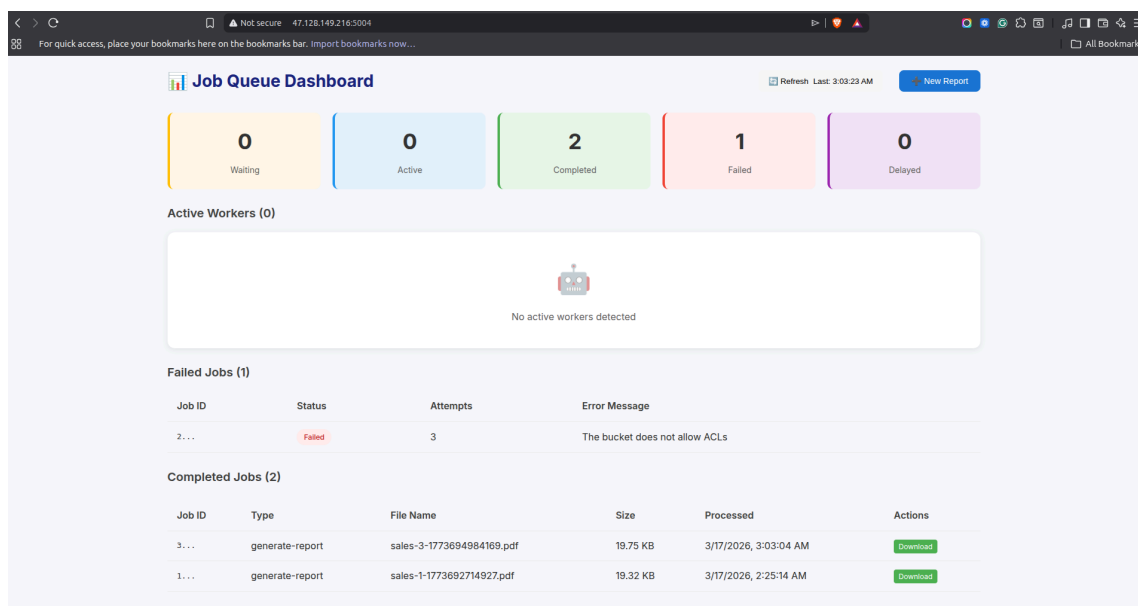
1. **Job Claim:** A worker locks a job from Redis.
2. **Rendering & PDF Generation:** Generates dynamic HTML and renders a PDF via Headless Chrome (Puppeteer).
3. **S3 Persistence:** The generated PDF is uploaded directly to AWS S3.
4. **Completion/Retry:** The worker releases the lock or automatically retries with exponential backoff upon failure.

4. Cloud Deployment & Infrastructure Snapshot

The system is deployed using **Pulumi** to manage networking, compute, and storage on AWS. Below is a snapshot of the provisioned resources in the AWS console:



A live snapshot of the dashboard analytics UI running on the provisioned AWS EC2 instance:



5. Key Strengths

- **Decoupled Architecture:** Highly available and easily scalable horizontally by adding more worker processes.
- **Robust Persistence:** S3 provides reliable, highly accessible, and fault-tolerant storage for generated artifacts.
- **Modern GitOps Pipeline:** Automated infrastructure provisioning with Pulumi and deployment using GitHub Actions limit downtime and human errors.